PROJECT TITLE: GreenTrace: Custodial Chain and Geospatial Mapping for DENR Seedling Reconciliation

Keywords: *Role-Based Access Control*

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CHAPTER 1: TECHNICAL DESCRIPTION & PROPOSAL

1.1 Background of the Study

Loggers that request cutting permits are mandated to supply 100 seedling replacement for every tree cut. Upon turn-over process, permittees shall provide the necessary amount of seedlings together with the letter sent to them. The receiving section of DENR is responsible for assuring the completeness of the turned-over seedlings and storing them in the nursery. Permittee details are recorded using logbook and Excel along with their compliance record. Agencies that request seedlings for tree planting are accompanied by NGP staff upon retrieval and planting. The details for tree planting are recorded e.g. amount of seedlings, agency, location, tree species. For monitoring, the NGP utilizes geotagging for extracting the exact coordinates of the tree planting site.

Among the issues that emerge in this process is data fragmentation due to separated and manually processed data records. Data gathering results revealed that the Monitoring and Enforcement Section encountered issues upon handling the logbook and the spreadsheet due to the fact that only one staff fully understands the record. Our system, GreenTrace, has a goal to provide a solution for this challenge by creating a unified system that includes critical processes involved in accordance with Uniform Replacement Ratio and National Greening Program.

1.2 Objectives (General and Specific)

- General Objective: *To design, develop, and deploy a web-based Monitoring System using the MERN stack for PENRO Marinduque that processes seedling replacement compliance and reforestation by the end of the current semester.*
- Specific Objectives: 1. *To implement a secure User Authentication module using JWT and bcryptjs, separating access between 'Admin', 'FRUU', 'MES', and 'NGP' roles.* 2. *To develop a RESTful API backend that processes dynamic CRUD operations for at least 3 distinct database collections.* 3. *To deploy the fully functional database, server, and client applications to cloud hosting platforms with zero critical build errors.*

1.3 Review of Related Literature and Systems

- ***Geospatial Monitoring & Precision Agriculture***

  Provide a systematic mapping on agricultural remote sensing techniques, showing how geographic information systems (GIS) and GPS are crucial in capturing field data and improving decision support systems in agriculture. Their review highlights trends in the use of remote sensing for parameter estimation, growth monitoring, and spatial analysis of crops, emphasizing the importance of geospatial data integration for precision management[1].

- ***GIS-Based Forestry Data Systems***

  GIS-based forestry data systems are widely used for land classification and environmental monitoring. These systems integrate spatial and attribute data to provide comprehensive analysis. They support mapping of forest resources and land use patterns. Additionally, they enable data sharing among government agencies. The use of GIS improves planning and resource management. These systems are also scalable and adaptable to different regions. However, they are primarily focused on macro-level data. They do not track individual seedlings or their movement across different stages. There is also no built-in mechanism for reconciliation of distributed resources. Furthermore, user roles and permissions are often limited. Therefore, enhancements are needed to support detailed tracking and accountability[2].

- ***Automation & Traceability for Seedling/Asset Monitoring***

  A real-time traceability system for agricultural products based on wireless networks that captures both movement and geolocation data from sensors to central servers, illustrating improved monitoring beyond isolated manual tracking methods[3].

- ***Smart Farming & Data Governance***

  surveys on smart farming systems emphasize challenges in data management, privacy, and integration of diverse sensors and monitoring technologies. This highlights the need for robust data structures and access control mechanisms to ensure reliability across multi-stakeholder environments[4].

- ***Seedling Tracking & Automated Monitoring***

  Systems that employ RFID and remote sensing have been proposed for greenhouse seedling management, allowing automated tracking of seedlings and enhancing traceability compared to manual record systems. This supports more reliable inventory tracking with linked transaction records[5].

- ***UAV-Based Seedling Mapping System***

The UAV-Based Seedling Mapping System utilizes unmanned aerial vehicles equipped with high-resolution cameras to capture detailed imagery of reforestation sites. These images are processed using advanced image analysis techniques to identify and locate individual seedlings with a high degree of spatial accuracy. The system significantly improves efficiency compared to traditional ground surveys, reducing labor and time requirements. It also enables monitoring of large and inaccessible areas, which is particularly useful in forest management. Additionally, spectral data analysis allows for the detection of vegetation health and growth patterns over time. The integration of geospatial data enhances decision-making for forestry managers. However, the system has limitations in distinguishing between naturally grown and artificially planted seedlings. It also lacks a transactional tracking mechanism to monitor the movement of seedlings from nurseries to planting sites. Furthermore, the system does not provide user-level access control or centralized database management. These limitations highlight the need for integrating UAV mapping with a custodial chain tracking system to improve seedling reconciliation processes[6].

- ***Remote Sensing-Based Forest Monitoring Systems***

  Remote sensing-based forest monitoring systems utilize satellite imagery and hyperspectral sensors to observe forest conditions over time. These systems are capable of detecting large-scale environmental changes such as deforestation, forest degradation, and vegetation health. They provide valuable data for policy-making and environmental conservation efforts. The use of multi-temporal datasets allows for trend analysis and forecasting. Additionally, cloud-based platforms enable data sharing among multiple stakeholders. The system is highly scalable and can cover vast geographic regions efficiently. However, its spatial resolution is often insufficient for monitoring individual seedlings. The data is also not always available in real-time due to satellite revisit intervals. Moreover, these systems lack integration with operational workflows such as seedling distribution and tracking. There is also limited capability for role-based access and localized data management. Therefore, while effective for macro-level monitoring, these systems are inadequate for detailed seedling reconciliation tasks[7].

- ***Machine Learning-Based Vegetation Mapping***

  Machine learning-based vegetation mapping systems apply algorithms such as random forests and convolutional neural networks to classify vegetation types. These systems improve classification accuracy by learning patterns from large datasets. They are widely used in environmental monitoring and land-use analysis. The automation of classification reduces human error and increases efficiency. Furthermore, these systems can integrate multiple data sources, including satellite and drone imagery. They are also adaptable to different ecological environments. However, the focus is primarily on vegetation classification rather than tracking individual seedlings. The models require large amounts of training data, which may not always be available. Additionally, the systems do not include features for tracking the lifecycle of seedlings. There is also no integration with database systems for monitoring distribution and accountability. Hence, they are limited in addressing the specific needs of seedling reconciliation systems[8].

- ***Forest Seedling Quality Monitoring System***

  The forest seedling quality monitoring system focuses on evaluating the health and viability of seedlings in nurseries. It uses standardized metrics to assess growth and quality. This ensures that only healthy seedlings are distributed for planting. The system improves the success rate of reforestation efforts. Additionally, it supports data recording and reporting. It also helps in maintaining quality control in nurseries. However, the system is limited to nursery-level monitoring. It does not track seedlings after distribution. There is no integration with geospatial mapping tools. It also lacks a centralized system for monitoring multiple locations. Therefore, it cannot support full lifecycle tracking of seedlings[9].

- ***Mobile Sensor-Based Seedling Monitoring***

  This system uses mobile sensors and GPS technology to monitor seedling growth and environmental conditions. It collects real-time data such as temperature, humidity, and soil quality. The system enhances accuracy in monitoring seedling development. It also allows field workers to collect data efficiently. Additionally, it supports mobile-based data entry. However, the system focuses only on data collection. It does not include centralized data processing or analytics. There is also no integration with GIS mapping. Furthermore, it lacks features for tracking seedling movement. Therefore, it cannot support reconciliation[10]

- ***Landsat-Based Seedling Monitoring***

  Landsat-based systems use satellite imagery to monitor vegetation changes over time. These systems are useful for detecting trends in forest growth. They provide historical data for analysis. Additionally, they support large-scale monitoring. However, their spatial resolution is low. They cannot detect individual seedlings. There is also a delay in data availability. Furthermore, they do not support real-time monitoring. These systems also lack integration with operational workflows. Therefore, they are not suitable for detailed seedling tracking[11].

- ***UAV and Deep Learning Tree Detection***

  This system combines UAV imagery with deep learning models for tree detection. It improves accuracy in identifying vegetation. The system automates detection processes. It also reduces manual effort. However, it focuses on detection only. It does not include tracking features. There is also no database integration. Furthermore, it lacks user management. Therefore, it is limited in application[12}.

- ***Geotagging and GPS Tracking Systems***

  Geotagging systems use GPS technology to attach location data to records. These systems are widely used in environmental monitoring. They help track the exact location of activities. Additionally, they improve data accuracy. GPS tracking supports mapping and analysis. However, these systems only provide location data. They do not include transactional tracking. There is also limited integration with databases. Furthermore, real-time tracking may be affected by signal issues. Thus, geotagging must be combined with other systems[13].

- ***Digital Mapping Systems for Environmental Planning***

  Digital mapping systems are widely used for environmental planning and resource management. They allow visualization of spatial data in interactive formats. These systems improve decision-making by providing clear geographic insights. They also support integration of multiple datasets. Additionally, digital maps can be updated in real-time. However, most systems focus on visualization only. They lack operational tracking features. They also do not manage transactional data. Furthermore, they are not designed for custodial chain monitoring. Thus, they need enhancement for forestry applications[14].

- ***Forest Resource Management Systems***

  Forest resource management systems are designed to support planning and monitoring of forest resources. They integrate spatial data and administrative information. These systems improve resource allocation efficiency. They also help in environmental conservation planning. However, they focus on macro-level management. They do not track seedlings individually. They also lack real-time updates. Furthermore, they do not include accountability mechanisms. Thus, they require enhancement for operational tracking[15],

1.4 Methodology & Solution

The project follows a practical system development approach that begins with requirements gathering from the approved project concept, then translates those requirements into modules, routes, database collections, and user interfaces. The solution uses a MERN architecture: React with Vite for the client interface, Node.js with Express for the REST API, and MongoDB with Mongoose for persistent storage.

The backend exposes versioned API endpoints under /api/v1. Authentication is handled through login and token verification, while authorization is enforced through strict permission checks per role. The frontend consumes these endpoints through protected routes and role-based navigation. CRUD workflows were planned for permits, seedlings, planting sites, and staff accounts, while reports are restricted to monitoring personnel.

Testing is intended to include route validation, permission denial checks, input validation, token checks, and manual API verification through Postman. Deployment planning includes a cloud-hosted backend and frontend, with environment variables used to manage secure configuration values.

1.5 Target Users / Beneficiaries

- Admin: *The Administrator serves as the role for assigned ICT staff of PENRO and their responsibility is to create and manage the account for users.*

- FRUU*: The FRUU initializes the workflow by registering the details of the issued permit which mainly includes permit number and amount of trees cut. This entry will allow the system to calculate the seedling debt of the corresponding permittee which shall be entered into the database in order to forward to MES.*

- MES: *The surrendered number of seedlings will be inputted to the seedling stocks table and categorized into Surrendered Seedlings. Furthermore, MES user can input the credentials of the permittee and import KML file for checking the geographical data and consider the planting site's alignment with the designated NGP sites.*
- NGP: *Their responsibility is to shift the appropriate number of seedlings to the corresponding tree planting record and supply the appropriate details required. In addition, they can utilize the system for uploading photo evidence of tree planting activity which will extract the geographic coordinates and automatically pin on map.*

1.6 Significance of the Study

The developed system would benefit the Provincial Environment and Natural Resources Office (PENRO) of Marinduque. The system will enable a convenient and easier way to monitor seedling replacement obligations and reforestation transparency. The system will enable real-time stock reconciliation instead of manually cross referencing paper records, thus, challenges with data fragmentation are eliminated through centralized database. As a result, the regional office will be provided with real-time and reliable data for more rigorous audit and monitor the reforestation targets that are based upon verified field data.

CHAPTER 2: MINIMUM VIABLE PRODUCT (MVP) DOCUMENTATION

2.1 System Architecture

The GreenTrace MVP follows a three-layer web architecture. The presentation layer is a React + Vite frontend that displays dashboards, forms, maps, and role-protected pages. The application layer is an Express API that validates requests, enforces permissions, and performs business logic. The data layer is MongoDB, accessed through Mongoose models for users, permits, seedlings, and planting sites.

A user signs in through the frontend, receives a JWT, and then accesses only the modules allowed by the assigned role. API requests are sent with bearer tokens. The backend validates the token, checks the role, and either grants access to data or returns an authorization error. This structure keeps permissions centralized and supports future deployment to online hosting services.

2.2 Tech Stack & Tools

- Frontend: React.js, Vite, React Router, Axios
- Backend: Node.js, Express.js
- Database: MongoDB Atlas, Mongoose
- Testing: Postman, Route validation, RBAC denial checks
- Deployment: Vercel / Render

2.3 Database Schema (Data Modeling)

- Users: *Fields include fullName, email, password, role, organizationUnit. Role enum includes ADMIN, FRUU, NGP, MES; password is hashed.*
- Permits: *Fields include permitNumber, location, treesCut, requiredSeedlings, status, issuedBy. The corresponding full name of the user fills the field for issuedBy.*
- Seedlings: *Fields include sourceName, species, qtyIn, qtyOut, balance, permit, plantingSite, recordedBy. The requiredSeedlings is linked to the qtyIn and qtyOut which indicates the amount of turned-over seedlings and planted seedlings, respectively.*
- Planting Sites*: Fields include locationName, municipality, coordinates, species, qtyPlanted, status, photos. This collection is connected to the plantingSite field in the Seedlings connection, specifically the plantingSite, species, and qtyPlanted.*

CHAPTER 3: REST API DESIGN DOCUMENTATION

3.1 Overview & Intent

- Base URL: https://greentrace-backend-eue1.onrender.com/api/v1
- Purpose: *The API exposes authentication, user management, permit tracking, seedling reconciliation, planting site monitoring, upload handling, and report generation endpoints for the GreenTrace system.*

3.2 Authentication & Authorization

- Strategy: *JSON Web Tokens (JWT) authentication is used in the system. After a successful login, the server returns a token. This token must be included in protected requests using the header.*
- Roles:

  ADMIN - manages staff accounts and roles

  FRU - manages permits

  NGP - manages seedlings and planting sites

  MES - manages permit, seedlings, planting sites, and reports

3.3 Endpoint Documentation

- Endpoint Name: Register NGP User
- Method & URL:[POST] https://greentrace-backend-eue1.onrender.com/api/v1/auth/register
- Authorization Required: No
- Request Format (Headers & Body):

```
JSON
{
  "fullName": "Juan Dela Cruz",
  "email": "ngp@denr.gov.ph",
  "password": "Ngp12345",
  "organizationUnit": "PENRO Marinduque"
}

Success Response (201 Created):
{
  "_id": "sample_user_id",
  "fullName": "Juan Dela Cruz",
  "email": "ngp@denr.gov.ph",
  "role": "NGP"
}

Error Response (400 Bad Request):
{
  "message": "Email already exists"
}
```

- Endpoint Name: User Login
- Method & URL:[POST] https://greentrace-backend-eue1.onrender.com/api/v1/auth/login
- Authorization Required: No
- Request Format (Headers & Body):

```
JSON
{
"email": "fru@denr.gov.ph",
"password": "Fru123"
}

Success Response (200 OK):
{
"token": "jwt_token_here",
"user": {
"_id": "sample_user_id",
"email": "fru@denr.gov.ph",
"role": "FRU"
}
}

Error Response (401 Unauthorized):
{
"message": "Invalid credentials"
```

```
}
```

- Endpoint Name: Get Current User
- Method & URL:[GET] https://greentrace-backend-eue1.onrender.com/api/v1/auth/me
- Authorization Required: Yes
- Request Format (Headers & Body):

```
JSON

Authorization: Bearer <token>

Success Response (200 OK):
{
"_id": "sample_user_id",
"fullName": "FRU Officer",
"email": "fru@denr.gov.ph",
"role": "FRU",
"organizationUnit": "DENR"
}

Error Response (401 Unauthorized):
{
"message": "Not authorized, no token"
}
```

- Endpoint Name: Get All Users
- Method & URL:[GET] https://greentrace-backend-eue1.onrender.com/api/v1/auth/users
- Authorization Required: Yes (ADMIN only)
- Request Format (Headers & Body):

```
No request body required



Success  Response(200 OK)

{
    "_id": "69d72e3a68b9dfcc7fff3283",
    "email": "newngp@denr.gov.ph",
```

```
    "fullName": "New NGP Officer",
    "role": "NGP",
    "createdAt": "2026-04-09T04:42:34.143Z",
    "updatedAt": "2026-04-09T04:42:34.143Z"
  },
  {
    "_id": "69d700ff9d7ec489efad77d2",
    "email": "mes@denr.gov.ph",
    "fullName": "chavez",
    "role": "MES",
    "organizationUnit": "DENR",
    "createdAt": "2026-04-09T01:29:35.963Z",
    "updatedAt": "2026-04-09T01:29:35.963Z"
  },
  {
    "_id": "69d7009e195785a6b0114ae9",
    "email": "admin@greentrace.com",
    "fullName": "Admin",
    "role": "ADMIN",
    "organizationUnit": "System Administration",
    "createdAt": "2026-04-09T01:27:58.313Z",
    "updatedAt": "2026-04-09T01:32:31.076Z"
  },
  {
    "_id": "69d6ff059d7ec489efad778e",
    "email": "ngp@denr.gov.ph",
    "fullName": "Joel",
    "role": "NGP",
    "organizationUnit": "DENR",
    "createdAt": "2026-04-09T01:21:09.583Z",
    "updatedAt": "2026-04-09T01:21:09.583Z"
  },
    {
    "_id": "69d310b57d074cdcc3d22073",
    "email": "fru@denr.gov.ph",
    "fullName": "FRU Officer",
    "role": "FRU",
    "organizationUnit": "DENR",
    "createdAt": "2026-04-06T01:47:33.315Z",
    "updatedAt": "2026-04-06T01:47:33.315Z"
  }
```


Error Response (403 Forbidden):


```
{
  "message": "Access denied. Admin only."
}
```


- Endpoint Name: Create Staff Account

- Method & URL:[POST] https://greentrace-backend-eue1.onrender.com/api/v1/auth/staff
- Authorization Required: Yes (ADMIN only)
- Request Format (Headers & Body):

JSON

```
{
  "fullName": "Pedro Reyes",
  "email": "mes@denr.gov.ph",
  "password": "Mes12345",
  "organizationUnit": "PENRO Marinduque",
  "role": "MES"
}
```

Success Response (201 Created):

```
{
  "_id": "sample_user_id_3",
  "fullName": "Pedro Reyes",
  "email": "mes@denr.gov.ph",
  "role": "MES",
  "organizationUnit": "PENRO Marinduque"
}
```

Error Response (400 Bad Request):

```
{
  "message": "Email already exists"
}
```

- Endpoint Name: Update User Role
- Method & URL:[PATCH] https://greentrace-backend-eue1.onrender.com/api/v1/users/:id/role
- Authorization Required: Yes (ADMIN only)
- Request Format (Headers & Body):

```
{
  "role": "FRU"
}
```

```
Success Response (200 OK):

{
"_id": "sample_user_id_3",
"fullName": "Pedro Reyes",
"email": "mes@denr.gov.ph",
"role": "FRU",
"organizationUnit": "PENRO Marinduque"
}
Error Response (400 Bad Request):
{
  "message": "Invalid role"
}
```

- Endpoint Name: Get All Permit
- Method & URL:[GET] https://greentrace-backend-eue1.onrender.com/api/v1/permits
- Authorization Required: Yes (FRU, MES)
- Request Format (Headers & Body):

```
JSON
Authorization: Bearer <token>

Success Response (200 OK):
[
{
"_id": "permit_id_1",
"permitNumber": "P-001",
"location": "Boac, Marinduque",
"treesCut": 5,
"requiredSeedlings": 500,
"status": "Active"
}
]

Error Response (403 Forbidden):
{
"message": "Access denied"
}
```

- Endpoint Name: Create Permit
- Method & URL:[POST] https://greentrace-backend-eue1.onrender.com/api/v1/permits
- Authorization Required: Yes (FRU, MES)
- Request Format (Headers & Body):

JSON
Authorization: Bearer <token>


```
{
"permitNumber": "P-002",
"location": "Mogpog, Marinduque",
"treesCut": 6,
"requiredSeedlings": 600,
"status": "Draft"
}
```


Success Response (201 Created):


```
{
"_id": "permit_id_2",
"permitNumber": "P-002",
"location": "Mogpog, Marinduque",
"treesCut": 4,
"requiredSeedlings": 400,
"status": "Draft"
}
```


Error Response (400 Bad Request):


```
{
"message": "Permit number is required"
}
```


- Endpoint Name: Get All Seedlings
- Method & URL:[GET] https://greentrace-backend-eue1.onrender.com/api/v1/seedlings
- Authorization Required: Yes (NGP, MES)
- Request Format (Headers & Body):

JSON

Authorization: Bearer <token>

Success Response (200 OK):


```
[
 {
   "_id": "seedling_id_1",
   "species": "Narra",
   "qtyIn": 1000,
   "qtyOut": 200,
   "balance": 800
```

```
 }
]

Error Response (403 Forbidden):
{
 "message": "Access denied"
}
```

- Endpoint Name: Get All Planting Sites
- Method & URL:[GET] https://greentrace-backend-eue1.onrender.com/api/v1/planting-sites
- Authorization Required: Yes (NGP, MES)
- Request Format (Headers & Body):

```
JSON
Authorization: Bearer <token>

Success Response (200 OK):
[
{
"_id": "site_id_1",
"locationName": "Mt. Malindig Reforestation Area",
"municipality": "Buenavista",
"coordinates": [121.95, 13.25],
"status": "Active"
}
]

Error Response (403 Forbidden):
{
"message": "Access denied"
}
```

- Endpoint Name:Get Reconciliation Reports
- Method & URL:[GET] https://greentrace-backend-eue1.onrender.com/api/v1/reports
- Authorization Required: Yes (MES only)
- Request Format (Headers & Body):

```
JSON

Request Format (Headers & Body):
Authorization: Bearer <token>

Success Response (200 OK):
{
"totalPermits": 10,
"totalSeedlings": 3000,
"activeSites": 4
}

Error Response (403 Forbidden):
{
"message": "Access denied"
}
```

3.4 Advanced API Features

- Pagination/Filtering/Sorting: *The current version of the API primarily uses standard CRUD endpoints. Some modules may support filtering or lookup endpoints such as permit options, site search, or geospatial search depending on the implemented route.*
- Versioning Strategy: *The API uses /api/v1 in the route path to support future updates without breaking older client integrations. Future major changes may be introduced under /api/v2.*

3.5 Testing Tools & Examples

- Tools Used: Postman was used for manual API testing, including authentication, CRUD validation, and role-based access checks.
- Sample URL/Fetch Example:

```
Javascript

fetch('https://greentrace-backend-eue1.onrender.com/api/v1/auth/me', {
  method: 'GET',
  headers: {
    'Authorization': 'Bearer <token>',
    'Content-Type': 'application/json'
  }
})
  .then(response => response.json())
```

CHAPTER 4: DEPLOYMENT & LINKS

4.1 Live Application Links

- Frontend Web Application URL: https://green-trace-mern.vercel.app
- Backend API Base URL: https://greentrace-backend-eue1.onrender.com
- GitHub Master Repository URL: https://github.com/joelp2002/GreenTrace-Mern

4.2 Testing Coverage Summary

- *Unit & Integration Tests:*

  *Basic backend testing was implemented using **Jest** and **Supertest**. The generated coverage report shows the executed test suites and the corresponding **statement**, **branch**, **function**, and **line** coverage of the backend modules. The test results show that **3 test suites passed successfully**, with a total of **12 passing tests**. These tests covered basic API health checking, invalid login handling, and protected route access denial.*

```
JoelParino@DESKTOP-5N85R9V MINGW64 ~/OneDriv
$ npm test

> greentrace-backend@1.0.0 test
> jest --detectOpenHandles --forceExit

 PASS  tests/auth.test.js (6.161 s)
 PASS  tests/protected.test.js
 PASS  tests/health.test.js

Test Suites: 3 passed, 3 total
Tests:       12 passed, 12 total
Snapshots:   0 total
Time:        10.554 s
Ran all test suites.
```

```
JoelParino@DESKTOP-5N85R9V MINGW64 ~/O
$ npx jest --coverage
  seedlingRoutes.js      |     100 |
  siteRoutes.js          |     100 |
  uploadRoutes.js        |     100 |
 backend/utils           |      40 |
  publicUrl.js           |      40 |
-----------------------|---------|---

Test Suites: 3 passed, 3 total
Tests:       12 passed, 12 total
Snapshots:   0 total
Time:        12.715 s
Ran all test suites.
```

```
JoelParino@DESKTOP-5N85R9V MINGW64 ~/OneDrive/Desktop/Final Output/greentrace(Mern)
$ npx jest --coverage
 PASS  tests/health.test.js (11.11 s)
 PASS  tests/protected.test.js (11.265 s)
 PASS  tests/auth.test.js (11.327 s)
-----------------------|---------|----------|---------|---------|-----------------
File                   | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-----------------------|---------|----------|---------|---------|-----------------
All files              |   40.08 |      9.6 |   15.55 |   41.48 |
 backend               |   87.87 |       20 |      75 |   87.87 |
  app.js               |   87.87 |       20 |      75 |   87.87 | 60-64
```

- End-to-End (E2E) Tests: End-to-End (E2E) Tests:

  Postman was used to test the complete workflow of the GreenTrace system from login up to accessing protected modules. The tests included admin login, staff account creation, role-based access to permits, seedlings, planting sites, and reports, as well as denied access for unauthorized roles. These tests confirmed that the system correctly uses JWT authentication and role-based access control in the deployed API.

APPENDICES

- Appendix A: Glossary of API Terms

  JSON (JavaScript Object Notation) – A lightweight data format used for sending and receiving structured information between the client and server.

  Endpoint – A specific URL route in an API that performs a particular function, such as login, retrieving records, or creating data.

  Payload – The data sent in the body of an HTTP request, usually in JSON format.

  Token – A generated authentication string returned after login and used to verify the identity of a user in protected requests.

  JWT (JSON Web Token) – A secure token format used for authentication and authorization in the system.

  Authorization Header – The HTTP header used to send the JWT token, usually in the format: Authorization: Bearer <token>.

  CRUD – Refers to the four basic database operations: Create, Read, Update, and Delete.

  Role-Based Access Control (RBAC) – A security mechanism that restricts access to modules and actions based on the assigned user role.

  API (Application Programming Interface) – A set of rules and endpoints that allows the frontend and backend to communicate with each other.

  HTTP Method – The request type used in an API call, such as GET, POST, PATCH, or DELETE.

- Appendix B: Acronyms & Abbreviations

  API – Application Programming Interface

  CRUD – Create, Read, Update, Delete

  DENR – Department of Environment and Natural Resources

  E2E – End-to-End

  EXIF – Exchangeable Image File Format

  FRU – Forest Resource Utilization Unit

  GIS – Geographic Information System

  GPS – Global Positioning System

JWT – JSON Web Token

KML – Keyhole Markup Language

MERN – MongoDB, Express.js, React.js, Node.js

MES – Monitoring and Enforcement Section

MVP – Minimum Viable Product

NGP – National Greening Program

PENRO – Provincial Environment and Natural Resources Office

RBAC – Role-Based Access Control

REST – Representational State Transfer

UI – User Interface

URL – Uniform Resource Locator